

# Ahamed Shakir M P

AI / ML Engineer · Full-Stack Developer

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## PROFILE

Computer Science & Engineering graduate (2026) specializing in AI/ML engineering and full-stack product development. Built end-to-end systems including an offline custom-AI investigation platform for Kerala Police, a real-time IoT safety network, and cross-platform mobile applications integrating hardware sensors. Comfortable across the full stack — Python ML backends, LLM integration, FastAPI services, React frontends, and React Native mobile apps. Seeking remote or international AI engineering positions.

## TECHNICAL SKILLS

**AI / ML:** LLMs, RAG Pipelines, PyTorch, NLP, Scikit-learn, Pandas, NumPy, ANN / Deep Learning, HuggingFace Transformers, Generative AI, Prompt Engineering

**Languages:** Python, JavaScript, TypeScript, Java, C

**Backend:** FastAPI, Node.js, Express.js, REST APIs

**Frontend / Mobile:** React, React Native, HTML, CSS

**Cloud & Data:** Firebase, MongoDB, Git, GitHub, Google Colab

**Hardware / IoT:** ESP32, GPS, IMU Sensors, BLE, WebSocket, microSD

**Design:** Figma, Wireframing, UX Prototyping, User Personas, Information Architecture

## PROJECTS

**FirAI — Kerala Police AI Investigation Assistant** *Python · PyTorch · FastAPI · React · PostgreSQL · Docker*

*A full-stack AI investigation platform built for Kerala Police under the Kerala AI Mission (K-AI), Kerala Startup Mission.*

- Built a 100% custom AI engine — a BiLSTM + Attention classifier (PyTorch) achieving 98.8% crime and severity classification accuracy with under 100ms CPU inference and a 3.3MB model — no third-party AI APIs, fully offline.
- Developed a custom NER pipeline, TextRank summarizer, and TF-IDF legal-section mapper, trained on Indian law texts (IPC, BNS, CrPC, BNSS) and 90+ real Kerala Police FIRs.
- Implemented multilingual FIR similarity search using Sentence-Transformer embeddings, modus operandi pattern detection across narratives, and Malayalam–English translation via Bhashini API.
- Delivered the full stack — FastAPI backend, React 19 dashboard, PostgreSQL, JWT authentication, Tesseract OCR PDF ingestion — deployed via Docker Compose; built for the Kerala AI Mission (K-AI) under Kerala Startup Mission (KSUM).

**Real-Time IoT Safety Monitoring Platform** *React Native · Firebase · ESP32 · Neo-6M GPS · MPU6050 · BLE · WebSocket*

*End-to-end IoT system integrating custom embedded firmware with a cross-platform mobile application.*

- Wrote and deployed firmware on ESP32 to simultaneously read GPS coordinates (Neo-6M module) and 6-axis motion data (MPU6050 IMU).
- Built real-time BLE and WebSocket communication pipeline streaming sensor data from hardware to React Native mobile app.
- Engineered accident detection using acceleration threshold analysis — automated SOS alerts and emergency contact notifications triggered on detection.
- Implemented geofencing, live map tracking, and offline-first microSD logging; Firebase Realtime DB syncs data for remote monitoring by family or emergency services.

**Disease Diagnosis Prediction — ANN** *Python · Scikit-learn · Pandas · NumPy*

*Multi-class medical diagnosis classifier using a trained Artificial Neural Network.*

- End-to-end pipeline: data cleaning (missing values, class imbalance, outliers), normalization, feature scaling, and categorical encoding.

- Designed multi-layer ANN; hyperparameter-tuned learning rate, batch size, and activation functions. Achieved 98%+ accuracy; evaluated with confusion matrix, F1, precision, and recall.

**Mobile App UX Design — μLearn Onboarding** *Figma · UX Prototyping · User Research*

- Designed 8-screen onboarding flow for the μLearn app (thousands of active users) — welcome through to dashboard — in Figma with full interactive prototype.
- Reduced onboarding steps by 30% through UX audit and consolidation; iterated on feedback from team leads before engineering handoff.

**EXPERIENCE**

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**Freelance Web Developer** · Sea Shark — UAE Client (Remote) 2026 – Present

- Designed and built a responsive business website for Sea Shark, a UAE-based client, using TypeScript and React — owning the project from requirements through deployment.
- Collaborated with an international client across time zones — managing communication, iterative revisions, and final handover independently.

**Campus Lead** · GTech μLearn (Kerala Government-backed student developer community) 2025 – 2026

- Led campus developer chapter; grew active participation by 40% through structured learning tracks and accountability systems.
- Organized 10+ technical workshops (AI, Web Dev, Cybersecurity, Open Source) reaching 100+ students per semester.
- Coordinated cross-institution collaboration with 10+ mentors to deliver hackathons and collaborative technical programs.
- Ranked #196 platform-wide on μLearn with 16,800+ karma earned across UI/UX, web development, and DevOps tracks.

**EDUCATION**

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**B.Tech — Computer Science & Engineering** 2022 – 2026

APJ Abdul Kalam Technological University, Kerala, India

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Computer Networks, Operating Systems, Embedded Systems, Software Engineering, OOP

**CERTIFICATIONS**

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| Microsoft Certified: Azure AI Fundamentals (AZ-900)                  | 2024       |
| Machine Learning & AI Internship — ICT Academy of Kerala             | 2024       |
| Emerging Technologies Internship (AI & Cloud) — Edunet × AICTE × IBM | 2024       |
| International Space Apps Challenge (NASA) — Participant              | 2023, 2024 |
| Introduction to Generative AI — Google                               | 2024       |
| Introduction to Large Language Models — Google                       | 2024       |
| Introduction to Responsible AI — Google                              | 2024       |
| Enterprise-grade AI & Data Science — IBM                             | 2024       |

**LANGUAGES**

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**English** — Professional working proficiency **Malayalam** — Native **Arabic** — Basic

**RESEARCH INTERESTS & MOTIVATION**

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Focused on applied AI research at the intersection of large language models and real-world deployment — specifically RAG architectures, semantic retrieval, Edge AI, and IoT-integrated intelligence systems. Interested in building products that address structural problems in public safety, healthcare access, and education.